

Final Presentation

Orthogonal Turning Digital Shadow

Team: Shadow Spindle

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SHADOW SPINDLE



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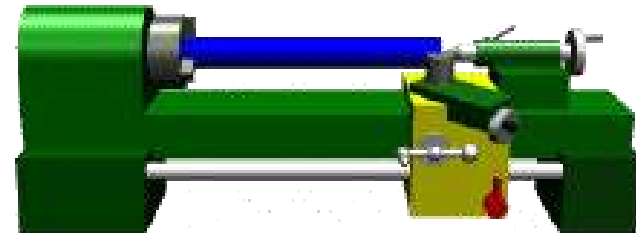
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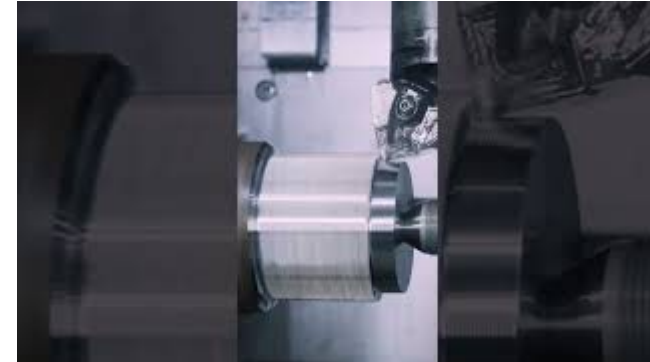


reference: <https://perfectcorporation.weebly.com/cnc-retrofitting.html>

Introduction

Project: Orthogonal Turning Digital Shadow

- Widely used in **aerospace machining**.
- Materials such as **titanium** and **nickel alloys** are **expensive**.
- Small machining deviations can damage the **workpiece surface**.



reference: <https://youtube.com/shorts/xSl2mpbWbc0?si=ucelCxapGOa7nTXd>

Motivation

- Maintain stable cutting conditions is critical to avoid defect parts.

Problem Statement

Problem: Vibrations outside the stability region cause **chatter**.

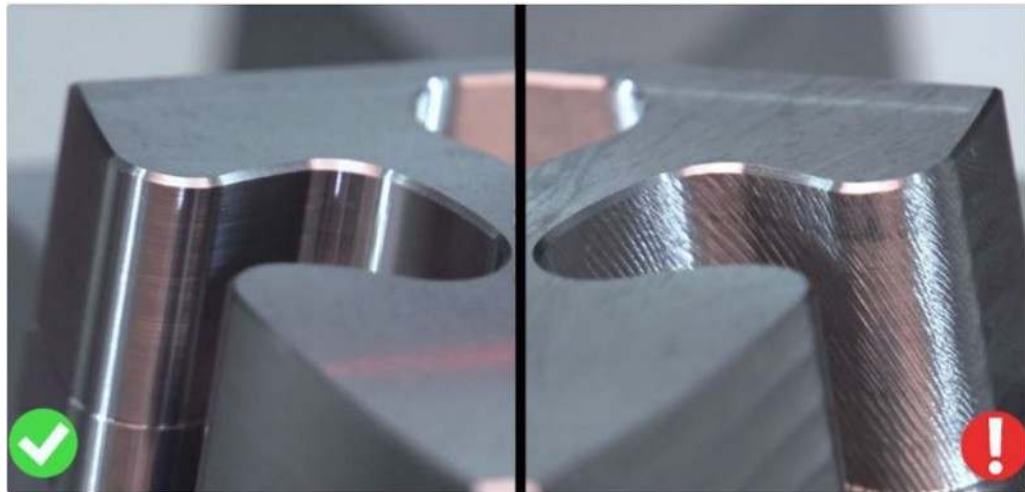
Effects of chatter:

- Poor surface quality.
- Tool wear.
- Scrap parts.
- #1 productivity offender in machining

Goal:

Develop a **Digital Shadow** to.

- monitor the machining process.
- keep the operation within the **stability region**.



reference: <https://www.dekmake.com/chatter-in-machining/>

Project Objectives

To develop a **microservice-based Digital Shadow** for monitoring an orthogonal turning process.

Project Components

- Collect machining signals (*acceleration, force, and sound*).
- Transform signals using **FFT**.
- Evaluate **cutting stability**.
- Visualize the process in **Godot**.
- Display indicators in a **dashboard**.
- Enable **prediction** and **reconfiguration**.



reference: [Why is Digital Twins Technology so Important to the Aviation Sector](#)

What was used to build the Digital Shadow



Container



Communication
between our services



Creation of the synthetic
data through a 1 degree
of freedom system
(1DOF)



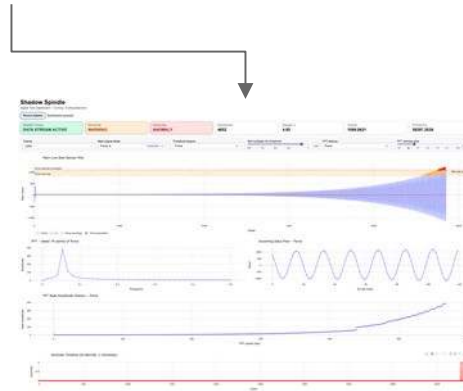
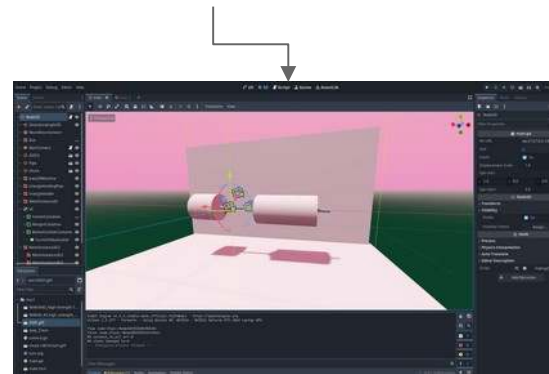
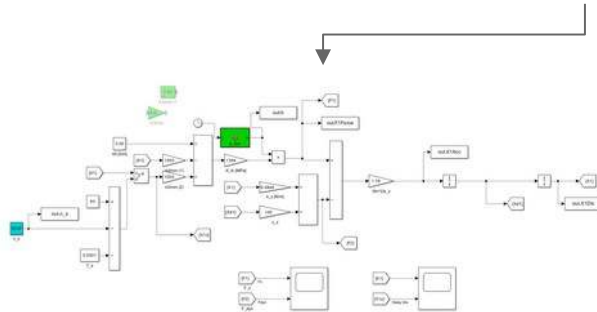
Visualization of the
system



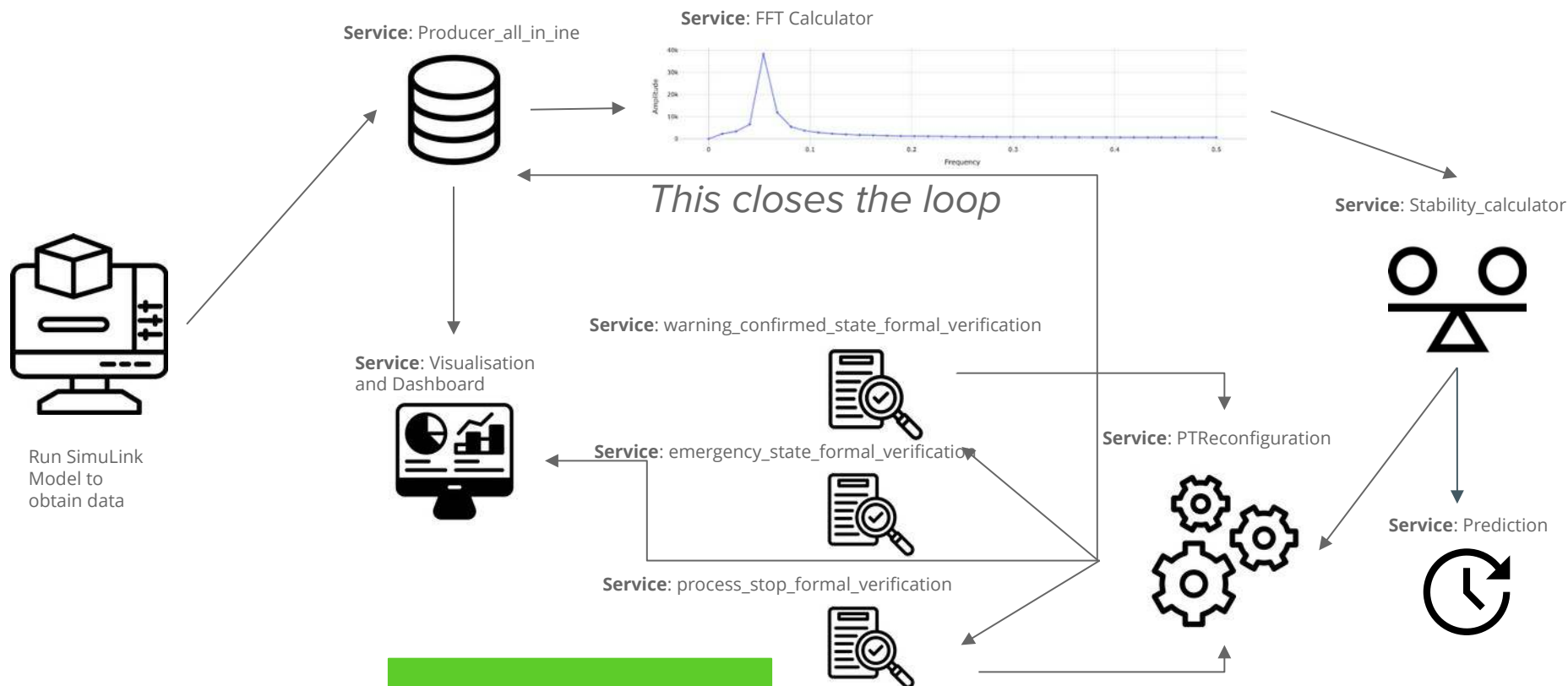
Used to create our
microservices and for
the Dashboard



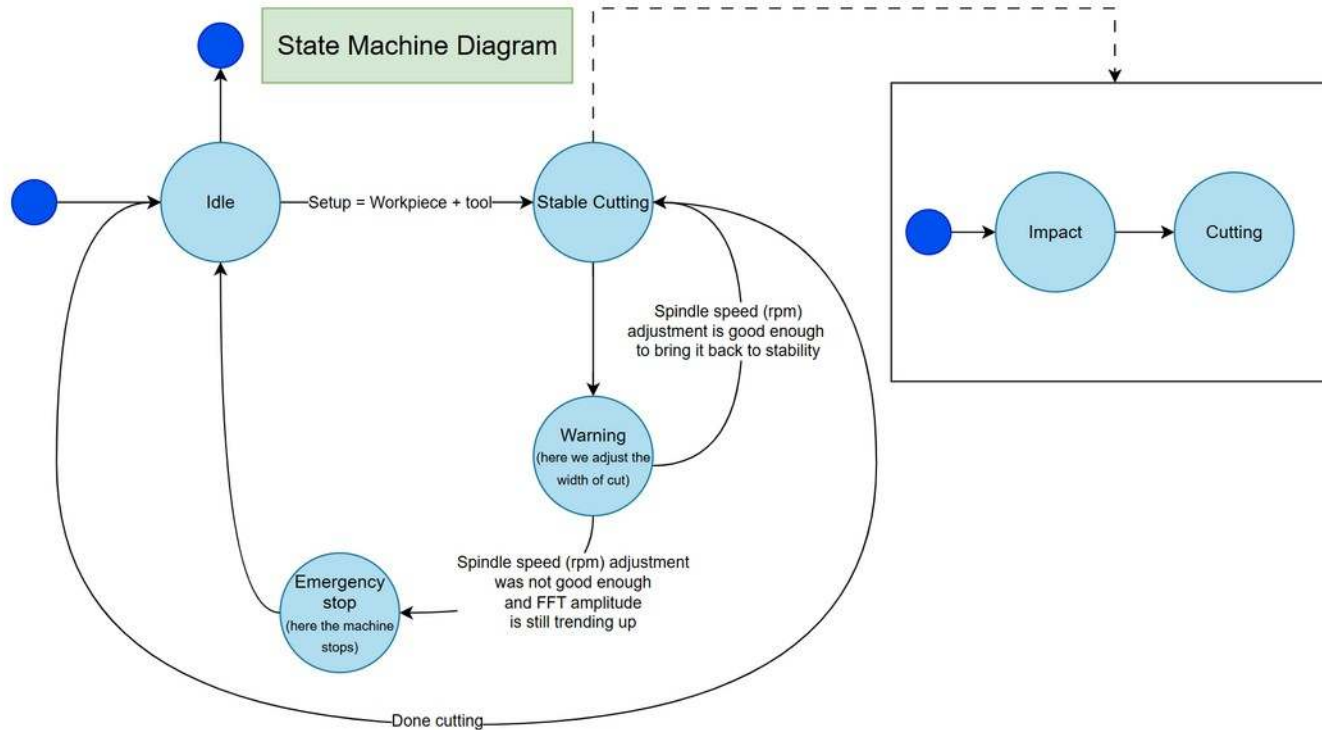
Used for versioning and
for teamwork



Our Digital Shadow Solution

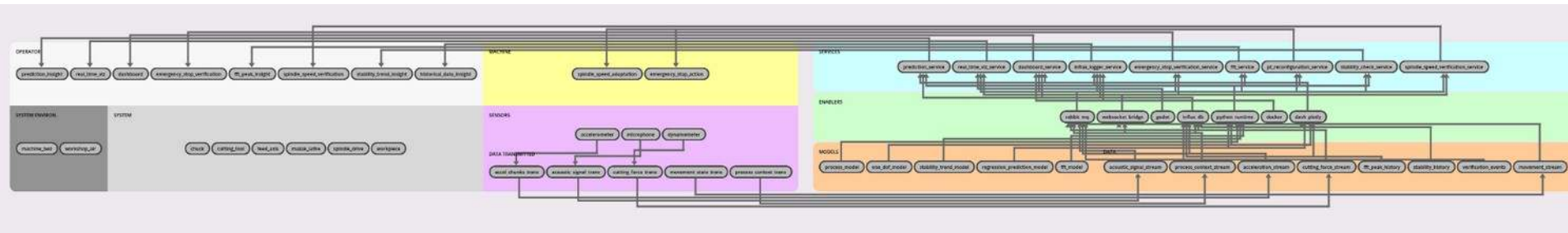


State Machines

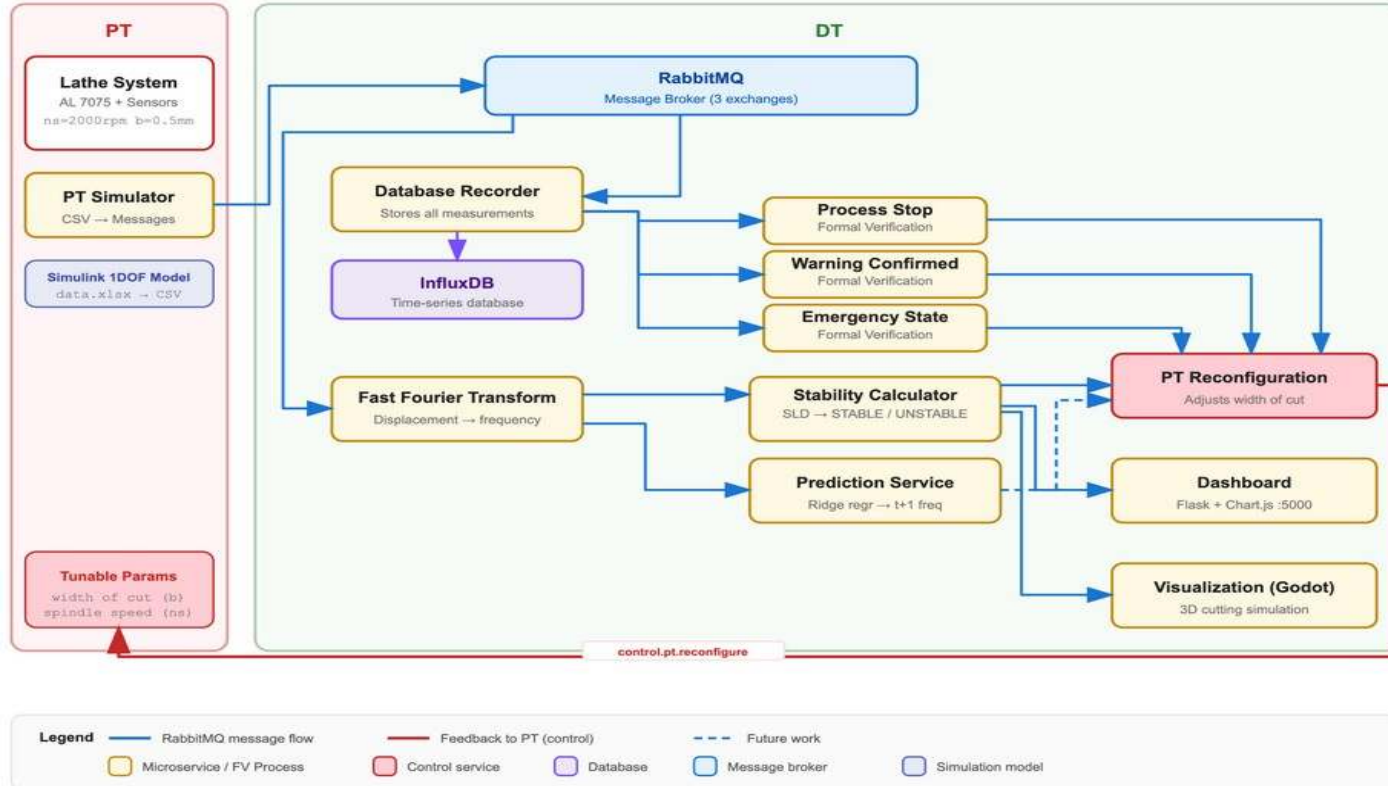


Constellation Diagram

- <https://haitham-alsatai.github.io/DTDF-Reporting/>



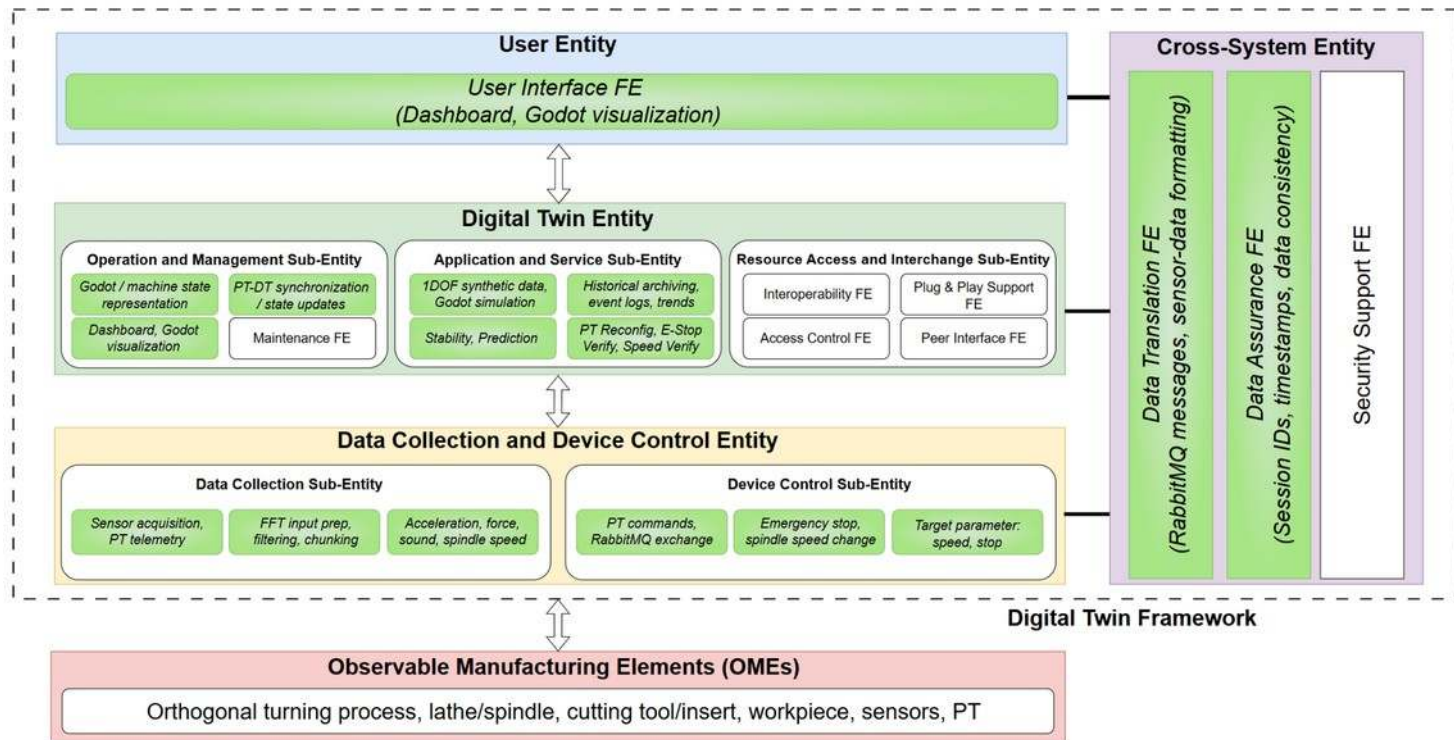
Our Microservices



Our Microservices

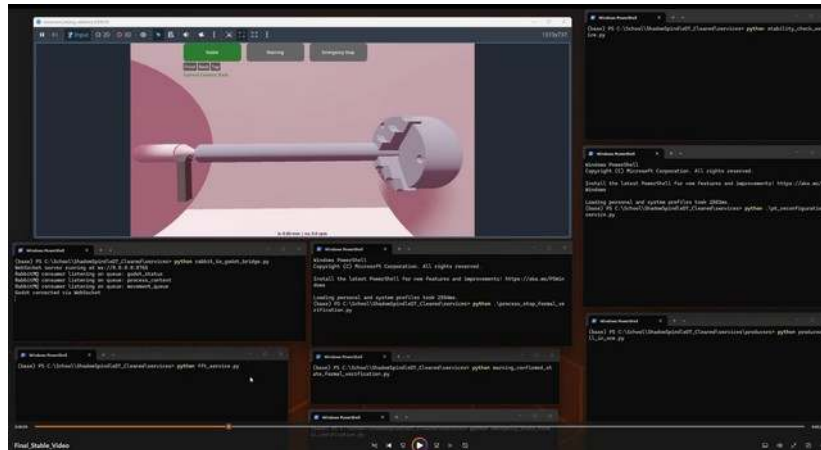
Service	Input	Output
<u>Data Producer / Simulation Service</u>	Pre-recorded simulation data from a spreadsheet file CSV / Excel simulation files	A continuous stream of sensor readings i.e accel_chunks (raw acceleration), movement_queue (tool position), process_context (b_mm , ns_rpm) — all published to RabbitMQ
Database Recording Service	All 6 RabbitMQ queues including accel_chunks, fft_peak, stability_trend, godot_status, movement_queue, process_context	InfluxDB (database) and Python dashboards (Dash and Plotly)
Database Query Service	InfluxDB time-series data	Pandas DataFrames; unified event replay timeline.
Frequency Calculator Service (FFT)	Acceleration measurements	Graph of amplitude of frequency and the coordinate with the highest frequency
<u>Stability Calculator Service</u>	Coordinate with the highest frequency	Stability status (STABLE /MARGINAL/UNSTABLE), critical width of cut (b_lim), stability ratio
Prediction Service	Coordinate with the highest frequency	Predicted Frequency at t+1, model accuracy
Dashboard Service	All analytics results + control commands	– (displays real-time web UI)
Visualization Service (Godot)	Stability status + sensor measurements	– renders 3D cutting simulation
<u>PT Reconfiguration Service</u>	Stability status	New width of cut value, reconfiguration command
Warning Formal Verification Service	Notification that a spindle speed reduction has been commanded	Confirmation that the 3D model has actually updated its spindle speed to match the new command — or an alert if it has not
Emergency Formal Verification Service	Notification that an emergency stop has been commanded	Confirmation that the 3D model has fully frozen and halted — or a repeated stop command if it has not responded correctly.
Process Stop Formal Verification	Notification that the machining process has completed successfully.	Confirmation that the spindle has reached 0 - or an alert if it has not stopped after process end.

ISO 23247

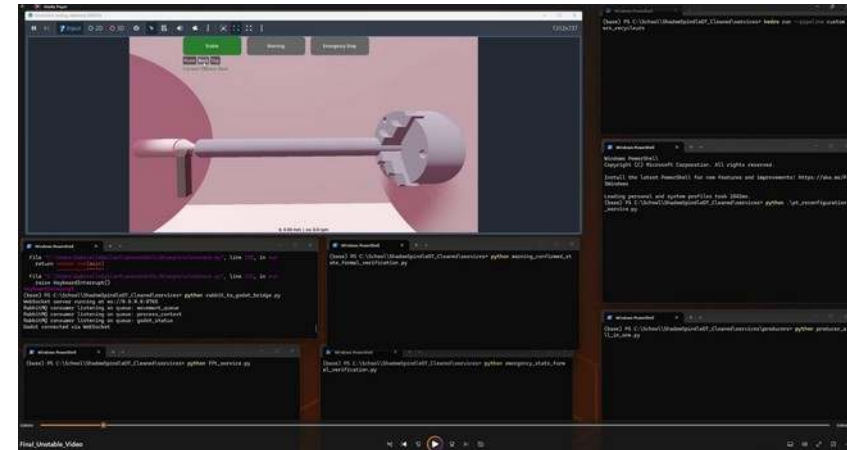


Demo of microservices (videos)

Stable



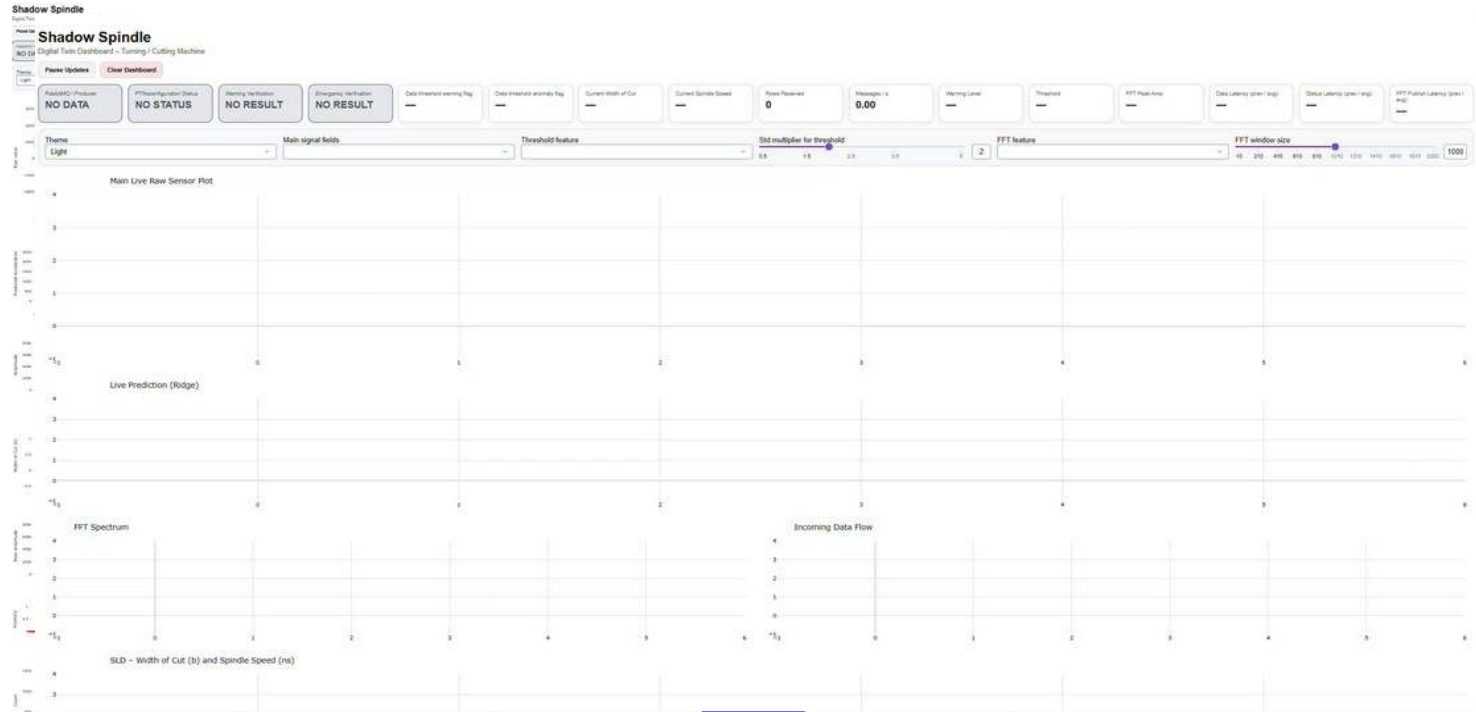
Unstable



States:

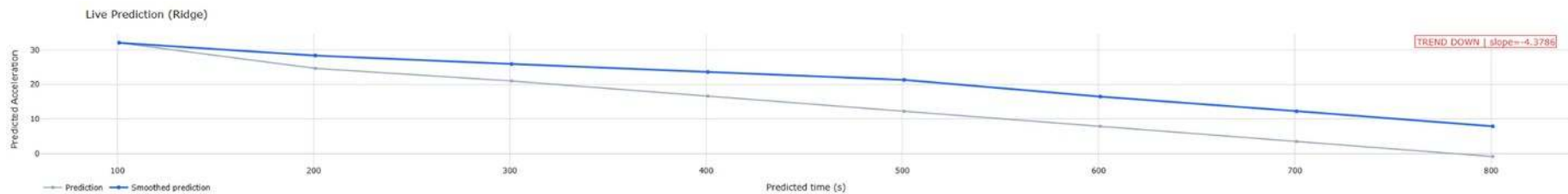
- **Stable:** no change.
- **Warning:** instability probable.
- **Warning:** instability confirmed reduce spindle speed by 10%.
- **Emergency stop:** stop all service and stop PT.

Demonstration of Dashboard (video)

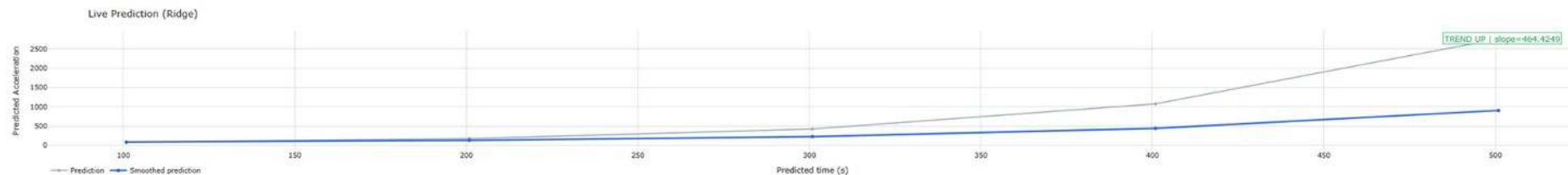


Demonstration of Prediction Service

● *Stable*



● *Unstable*



Dashboard Output

- *Snapshot*
- *Additional insight*
- *Data can be used by operator to make a decision*

```
Dashboard Snapshot | 2026-04-04 03:00:55

Stream State      : DATA STREAM ACTIVE
PT Status        : EMERGENCY STOP
Warning Verification : -
Emergency Verification : -
Warning Flag      : WARNING
Anomaly Flag      : ANOMALY
Width of Cut (b)  : 0.4700
Spindle Speed (ns) : 1800.0000
Rows Received     : 5000
Messages / s      : 0.9852
Warning Level     : 609.1662
Threshold         : 1217.8416
FFT Feature       : Acceleration
FFT Window        : 1000
FFT Peak Amp      : 855184.2362
Data Latency (ms) : 10.6461
Status Latency (ms) : 1.5280
FFT Publish Latency(ms): 1315.7182

Prediction
Predicted Time    : 500.9000
Predicted Value   : 2760.9204
Smoothed Prediction : 903.2208
Prediction Slope  : 464.4249
Prediction Trend  : TREND UP

Latest Rows
[
  {
    "timestamp": "2026-04-04 03:00:50.450218",
    "force": 2579.168696528211,
    "rpm": 1800.0,
    "b": 0.46999999999999998,
```

Results

- We created a **Digital Shadow** that monitor process **stability** during cutting using FFT.
- We worked closely with a **domain expert** to ensure the relevance and alignment of the Digital Shadow. (**Validation**)
- We created a **microservice architecture** that is scalable and maintainable.
- We created a **visualization** and **dashboard** that allow the domain expert to easily gain insight on what is going on in real-time.

Challenges

- *Multidisciplinary work (technical and domain expert required)*
- *Obtaining synthetic data (1DOF in Simulink)*
- *Determining the correct architecture and connecting everything together*

Lessons Learned

- *Importance of building the DS/DT alongside domain experts (validation)*
- *Integrating the standards in our design phase*
- *Existing tools (**RabbitMQ**, **Godot**, **Dash** in **python**, **InfluxDB**) available and determining which one is best suited for our project needs*

Limitations and Future Work

Limitations

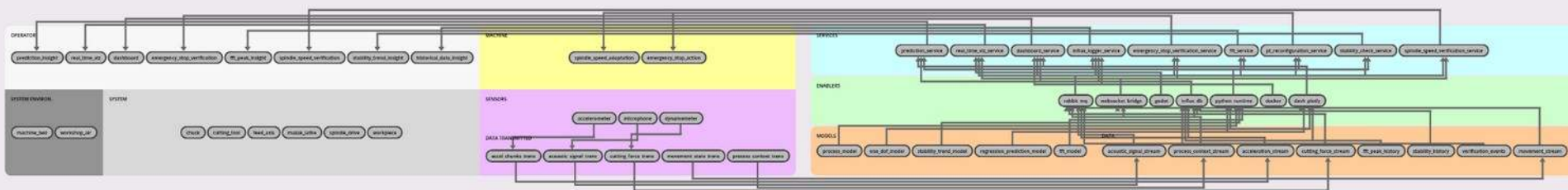
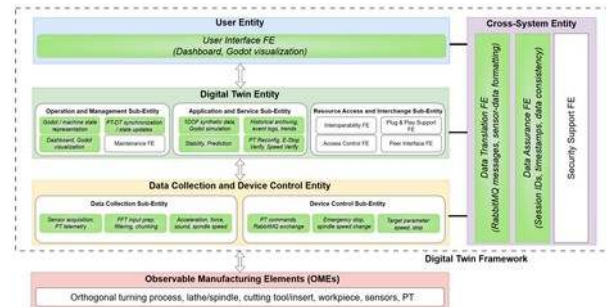
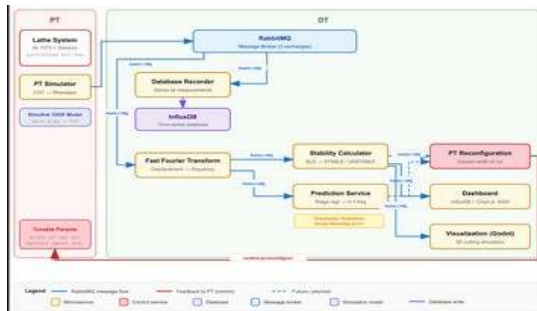
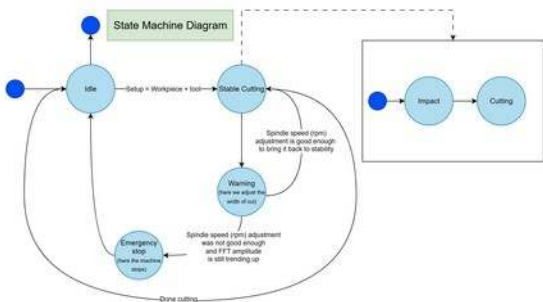
- *Only works for a very specific case of turning with no concept of resonance.*
- *Prediction service is only used for monitoring purposes*

Future Work

- *Make the user decide whether to use they prediction service instead of the stability service*
- *Test the DS architecture of a real turning machine*
- *Generalize the architecture*

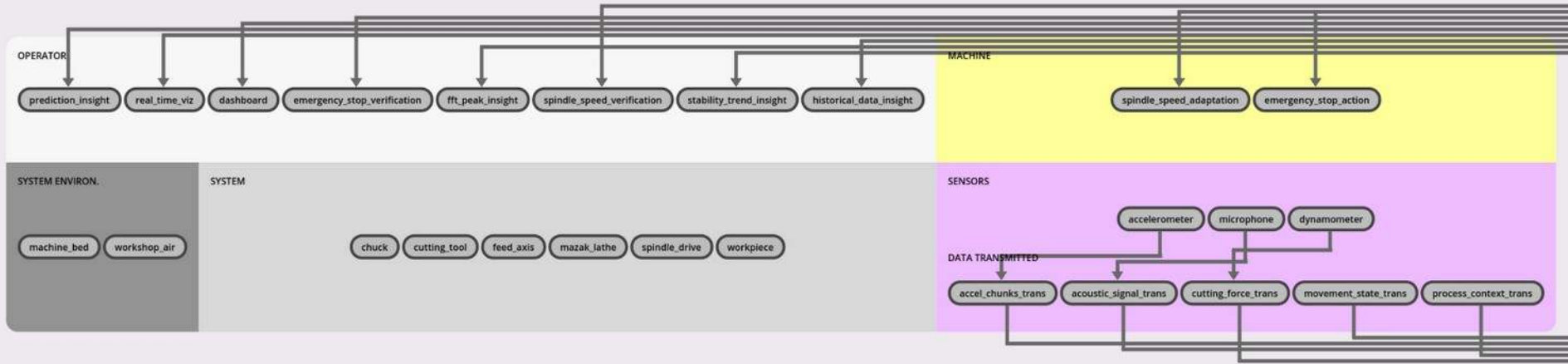
Thank you!

- Digital Shadow to maintain stability during orthogonal cutting
- Microservice architecture



Appendix - Constellation Diagram

● PT



Appendix - Constellation Diagram

- *Digital Shadow*

